

Sensor-based and Context-aware Interaction

Sensor-based interaction uses hardware sensors like GPS, accelerometer, camera, and microphone to interact with users. Context-aware interaction adapts system behavior based on user context such as location, time, activity, or environment. Example: Google Maps changing route suggestions based on traffic and location.

Ubiquitous Computing

Ubiquitous computing means computing is available everywhere and integrated into daily life. Applications include smart homes, wearable devices, healthcare monitoring, and smart cities.

Five Stage Search Framework

The five stages are: Formulation, Initiation, Review of results, Refinement, and Use. It explains how users search, evaluate results, and refine queries in HCI systems.

Pattern Recognition in HCI

Pattern recognition helps systems understand gestures, speech, handwriting, and facial expressions. It improves natural interaction, accessibility, and intelligent interfaces.

Machine Learning in HCI

Machine learning enables personalization, recommendation systems, adaptive interfaces, and predictive behavior in HCI.